

THE BEAUTY OF BACKPROPAGATION

A Pure & Comprehensive Chain Rule Blueprint (Bias-Free)

1. Loss Function (Categorical Cross-Entropy)

$$L = - \sum_{j=1}^K y_j \cdot \ln(p_j)$$

2. Pure Weight Gradient Chain Rule

$$\frac{\partial L}{\partial w_{km}^{(l-1)}} = \sum_j \sum_i \left(\frac{\partial L}{\partial p_j} \cdot \frac{\partial p_j}{\partial z_j^{logits}} \cdot \frac{\partial z_j^{logits}}{\partial h_i^{(l)}} \cdot \frac{\partial h_i^{(l)}}{\partial z_i^{(l)}} \cdot \frac{\partial z_i^{(l)}}{\partial z_k^{(l-1)}} \cdot \frac{\partial z_k^{(l-1)}}{\partial w_{km}^{(l-1)}} \right)$$

3. Layer & Index Reference System

Component / Layer	Indices Used	Description & Signals
Output Loss (L)	-	Categorical Cross-Entropy objective function.
Softmax / Logits	j	z_j^{logits} : raw output, p_j : probability after Softmax.
Upper Hidden Layer (l)	i	$z_i^{(l)}$: linear input, $h_i^{(l)}$: activated output (σ).
Lower Hidden Layer (l-1)	k	$z_k^{(l-1)}$: linear input, $h_k^{(l-1)}$: activated output (σ).
Prior Layer / Input (l-2)	m	$h_m^{(l-2)}$: output signal feeding into layer (l-1).
Network Input (x)	n	x_n : initial features entering the network.

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4. Pure Input Gradient Chain Rule

$$\frac{\partial L}{\partial z_i^{(l)}} \cdot \frac{\partial x_n}{\partial h_k^{(l-1)}} = \sum_j \left(\frac{\partial L}{\partial p_j} \cdot \frac{\partial p_j}{\partial z_j^{logits}} \cdot \frac{\partial z_j^{logits}}{\partial h_i^{(l)}} \cdot \frac{\partial h_i^{(l)}}{\partial z_i^{(l)}} \cdot \frac{\partial z_i^{(l)}}{\partial h_k^{(l-1)}} \cdot \frac{\partial h_k^{(l-1)}}{\partial z_k^{(l-1)}} \cdot \frac{\partial z_k^{(l-1)}}{\partial h_m^{(l-2)}} \cdot \frac{\partial h_m^{(l-2)}}{\partial z_m^{(l-2)}} \cdot \frac{\partial z_m^{(l-2)}}{\partial x_n} \right)$$

5. Side Analytical Component Breakdown

Derivative Member	Analytical Solution	Meaning
$\frac{\partial L}{\partial z_j^{logits}}$	$p_j - y_j$	Loss + Softmax simplified error vector.
$\frac{\partial z_j^{logits}}{\partial h_i^{(l)}}$	$w_{ji}^{lm_head}$	Weights mapping to logits layer.
$\frac{\partial h_i^{(l)}}{\partial z_i^{(l)}}$	$\sigma'(z_i^{(l)})$	Activation function valve at layer l.
$\frac{\partial z_i^{(l)}}{\partial h_k^{(l-1)}}$	$w_{ik}^{(l)}$	Weights connecting layer (l-1) to (l).
$\frac{\partial h_k^{(l-1)}}{\partial z_k^{(l-1)}}$	$\sigma'(z_k^{(l-1)})$	Activation function valve at layer l-1.
$\frac{\partial z_k^{(l-1)}}{\partial w_{km}^{(l-1)}}$	$h_m^{(l-2)}$	The incoming input signal from layer l-2.

6. Parameter & Input Modification Rules (SGD)

Weight Optimization	Input/Feature Optimization
$w_{km}^{(l-1)} \leftarrow w_{km}^{(l-1)} - \eta \cdot (\partial L / \partial w_{km}^{(l-1)})$	$x_n \leftarrow x_n - \eta \cdot (\partial L / \partial x_n)$